VTTSI-Chat: Real-Time Intersection Safety Monitoring with LLM-Powered Natural Language Explanation

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***Abstract*—We present VTTSI-Chat, a cloud-native system that pairs the Virginia Tech Transportation Safety Index (VTTSI)—a hybrid Real-Time Safety Index (RT–SI) and CRITIC-weighted MCDM pipeline—with a SafetyChat conversational module powered by a tool-augmented GPT-4o. VTTSI fuses seven years of VDOT crash records via Empirical Bayes stabilization with live connected-vehicle telemetry, producing interpretable 0–100 safety scores every 15 minutes. SafetyChat exposes six typed API tools to the LLM, grounding all natural language responses in live data and enabling TMC operators, planners, and emergency responders to query causal safety narratives through ordinary conversation. A live deployment at https://cs6604-trafficsafety-180117512369.europe-west1.run.app demonstrates four end-to-end use cases across monitored intersections in Falls Church, VA.**

I. Introduction

Real-time intersection safety data are increasingly available through connected-vehicle infrastructure, yet the gap between a numerical safety score and actionable operational insight remains large. Transportation Management Center (TMC) operators must interpret multi-dimensional indices under time pressure, often without the domain expertise to distinguish a volume-driven risk spike from a behavioral anomaly. Existing traffic dashboards present data but do not explain causality or suggest actions.

Recent work on tool-augmented LLMs demonstrates that function-calling models can reason over structured numerical data and produce accurate, grounded explanations [9, 7]. Within transportation, LLM efforts have targeted signal control [4] and AV motion planning [5, 12], but *safety-index explanation* for infrastructure operators remains unaddressed.

VTTSI-Chat addresses this gap. It extends the existing VTTSI backend—a hybrid RT–SI + MCDM scoring pipeline running on Google Cloud Run—with a SafetyChat module whose LLM issues structured function calls to live VTTSI endpoints, constructs evidence-grounded narratives, and delivers them through a conversational Streamlit interface.

Contribution. VTTSI-Chat is the first system to combine (1) a hybrid EB-stabilized crash index with real-time MCDM scoring, and (2) a tool-augmented LLM that generates causal safety narratives for TMC operators via natural language dialogue.

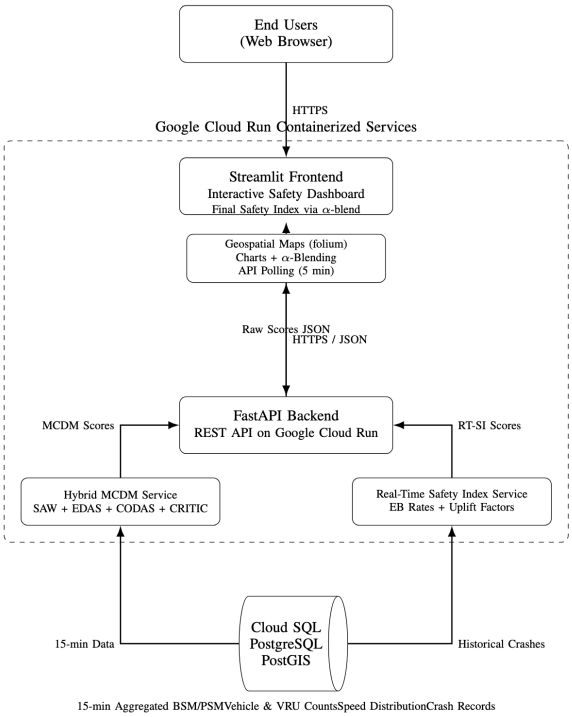

Fig. 1. VTTSI cloud architecture. The SafetyChat LLM module (dashed border) is the new contribution of this demo paper.

II. VTTSI System Overview

The VTTSI backend (FastAPI, Google Cloud Run) computes two complementary safety indices every 15 minutes from connected-vehicle telemetry streamed via the VTTI Trino system and stored in Cloud SQL PostgreSQL.

A. Real-Time Safety Index (RT–SI)

RT–SI anchors intersection risk to a severity-weighted Empirical Bayes (EB) baseline derived from 2017–2024 VDOT

crash records. The historical crash rate is:

$$r_i = \frac{\sum_s w_s C_{i,s}}{E_i}, \quad w_{\text{Fatal}} = 10,\ w_{\text{Inj}} = 3,\ w_{\text{PDO}} = 1, \quad (1)$$

where $C_{i,s}$ are crash counts by severity and E_i is vehicle exposure. The EB stabilized rate is:

$$\hat{r}^{\text{EB}}_{i,t}(\lambda) = \frac{Y_{i,t} + \lambda\, r_0}{1+\lambda}, \quad \lambda^\star = 10{,}000, \quad (2)$$

where r_0 is the global mean rate across all site-bins (2017–2024) and $\lambda^\star$ is chosen by minimizing Poisson log-loss on a 2025 hold-out set. Real-time uplift factors—bounded to $[0,1]$—adjust this baseline for speed deficit (F^{spd}), speed variance (F^{var}), and VRU–vehicle conflict exposure (F^{conf}):

$$U_{i,t} = 1 + \beta_1 F^{\text{spd}}_{i,t} + \beta_2 F^{\text{var}}_{i,t} + \beta_3 F^{\text{conf}}_{i,t}. \quad (3)$$

VRU and vehicle sub-indices are multiplied by $U_{i,t}$ and the EB baseline, combined with weights ω_{vru} and ω_{veh}, and min-max scaled to produce $\text{SI}^{\text{RT}} \in [0,100]$.

B. MCDM Safety Index

A CRITIC-weighted multi-criteria pipeline [1] constructs a decision matrix over five real-time criteria (vehicle count, VRU count, average speed, speed variance, incident count) for each intersection–time-bin pair. CRITIC criterion weights are derived from normalized standard deviations and inter-criterion Pearson correlations, recomputed every 24 hours. Three MCDM scoring methods—SAW, EDAS, and CODAS—are each computed and scaled to $[0,100]$; a second-pass CRITIC aggregation yields the hybrid index $\text{SI}^{\text{MCDM}} \in [0,100]$ (higher = higher risk).

C. Blended Final Index

$$\text{SI}^{\text{Final}}_{i,t} = \alpha\, \text{SI}^{\text{RT}}_{i,t} + (1-\alpha)\, \text{SI}^{\text{MCDM}}_{i,t}, \quad \alpha = 0.7 \text{ (default)}. \quad (4)$$

Scores map to risk levels: 0–40 = low, 41–70 = moderate, 71–100 = high. The α parameter is user-adjustable to shift emphasis between historical risk and instantaneous anomaly detection.

III. SafetyChat: LLM-Powered Explanation Module

A. Design

VTTSI already answers *what* the score is; SafetyChat answers *why* and *what to do*. We adopt the tool-augmented LLM pattern [9]: GPT-4o is equipped with typed function calls that map to VTTSI REST endpoints, ensuring every numerical claim in a generated narrative is grounded in live API data. An agentic loop (up to six iterations) allows the model to chain tool calls—for example, first ranking all intersections and then drilling into component breakdowns for the top-ranked sites.

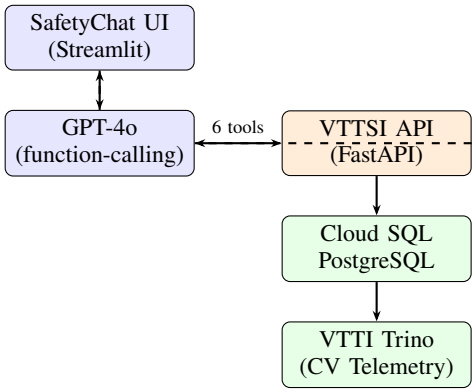

Fig. 2. SafetyChat data flow. Solid = request path; dashed = response path.

TABLE I
VTTSI-Chat vs. Related Systems

System	Hybrid Index	Real-Time Telemetry	LLM Interface	Causal Narrative
Google / Apple Maps	✗	✓	✗	✗
Waycare / ATSPM	✗	✓	✗	✗
TrafficGPT [11]	✗	✓	✓	✗
LLMLight [4]	✗	✓	✓	✗
GPT-Driver [5]	✗	✓	✓	✗
VTTSI-Chat (ours)	✓	✓	✓	✓

B. Tool Definitions

SafetyChat exposes six tools via the OpenAI function-calling protocol [7]:

1) `get_safety_score(intersection, target_time)` — returns RT–SI, MCDM, and blended scores with top uplift factors; `target_time` enables historical point-in-time queries.
2) `get_component_breakdown(intersection, target_time)` — returns CRITIC-weighted criterion values for a specific 15-minute bin.
3) `get_historical_baseline(intersection)` — returns EB prior, 7-year VDOT crash severity breakdown, and $\lambda^\star$.
4) `compare_intersections(metric, alpha)` — ranks all monitored intersections by any index component.
5) `get_trend_data(intersection, days_back)` — returns a 7–30 day MCDM score trend with direction indicator and up to 20 sampled time-bin snapshots.
6) `run_sql_query(query)` — executes any read-only SELECT against the live database for ad-hoc exploration; guarded against write operations.

C. Differentiation

Table I positions VTTSI-Chat against related systems. Existing LLM+traffic systems target signal control [4, 11] or AV motion planning [5, 12]; none pair a hybrid crash–telemetry safety index with operator-facing causal narratives.

TABLE II
SUMMARY OF DEMONSTRATION USE CASES

Use Case	Key Tool Calls	User
Morning briefing	compare + breakdown	TMC Operator
Causal explanation	score(t) + breakdown(t)	Planner
Responder routing	score × 2	Dispatcher
Trend engagement	get_trend_data(30d)	Official/Public

IV. DEMONSTRATION

The live demo runs against the production VTTSI deployment on Google Cloud Run, monitoring eight instrumented intersections in Falls Church, VA. Attendees interact with the SafetyChat interface directly. Four use cases are demonstrated.

A. Use Case 1: TMC Operator Morning Briefing

Query: "Give me a morning safety briefing for all intersections."
Action: SafetyChat calls `compare_intersections` to rank all sites by blended score, then `get_component_breakdown` for the two highest-ranked intersections. The response identifies peak-risk windows, dominant criteria (e.g., speed variance or elevated VRU counts), and compares current scores to the 7-day trend—synthesizing data from multiple API calls into a single situational-awareness narrative in under five seconds.

B. Use Case 2: Causal Safety Explanation for Planners

Query: "Why did E. Broad–N. Washington score above 70 yesterday at 5 PM?"
Action: SafetyChat passes `target_time` (`2025-11-20T17:00`) to `get_safety_score` and `get_component_breakdown`, identifies which criteria were in the 90th percentile for that bin, and traces the elevated score to concurrent peak vehicle volume and incident activity, consistent with the intersection's VDOT crash history.

C. Use Case 3: Emergency Responder Routing

Query: "Which of Glebe–Potomac or Birch–W. Broad has lower current risk for a priority crossing?"
Action: SafetyChat calls `get_safety_score` for both intersections, compares RT–SI, MCDM, and blended indices in real time, and recommends the lower-risk route with reasoning grounded in current VRU counts and speed variance.

D. Use Case 4: Public Stakeholder Trend Engagement

Query: "Is the Glebe–Potomac intersection getting safer over time?"
Action: SafetyChat calls `get_trend_data` with a 30-day window, reports the trend direction and mean/max/min MCDM scores, and translates index values into plain-language risk levels (e.g., "The score averaged 39.9 over 30 days—low risk, with an improving trend").

V. RELATED WORK

EB-stabilized crash models [8, 6] and MCDM-based indices [1, 2] provide the theoretical foundation for VTTSI's dual scoring pipeline. Surrogate safety indicators [10, 3] complement crash data with real-time conflict proxies, but lack a unified aggregation framework. In the LLM domain, TrafficGPT [11] targets traffic flow analysis, LLMLight [4] handles signal control, and GPT-Driver / DriveVLM [5, 12] operate at the AV layer—none provide a safety-index explanation interface for infrastructure operators. VTTSI-Chat uniquely combines (i) a validated hybrid index fusing long-term crash history with 15-minute telemetry, (ii) a tool-augmented LLM grounded in live API data, and (iii) causal narratives traceable to raw sensor signals.

VI. CONCLUSION

VTTSI-Chat is a live, cloud-native intersection safety system that closes the gap between quantitative safety scoring and human-readable situational awareness. By grounding all LLM outputs in six typed API tool calls, SafetyChat delivers causally grounded, hallucination-resistant safety narratives to TMC operators, planners, emergency responders, and the public through natural language dialogue. Future work will integrate the Agentic Traffic Safety Knowledge Graph (TS–KG) as an additional tool, enabling the LLM to reason over intersection geometry, conflict topology, and causal trajectory motifs.

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